

JWST/MIRI High-contrast Imaging of the η Corvi Exozodiacal Dust

Thomas Nguyen[1], Schuyler Wolff[1]

[1] Department of Astronomy & Steward Observatory, The University of Arizona, Tucson, AZ

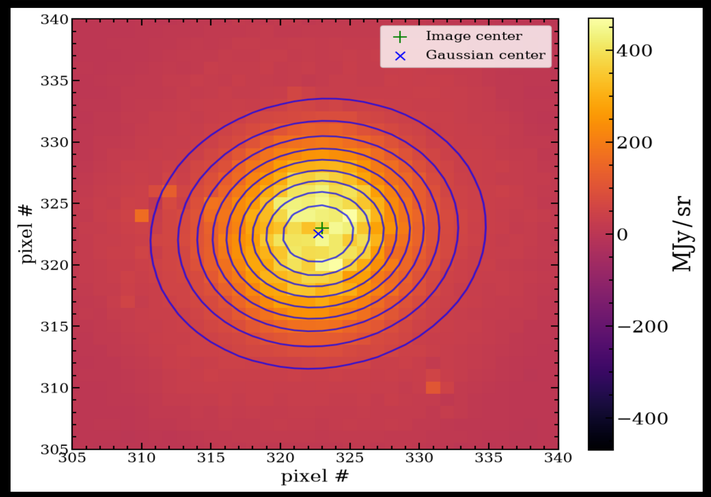


Overview

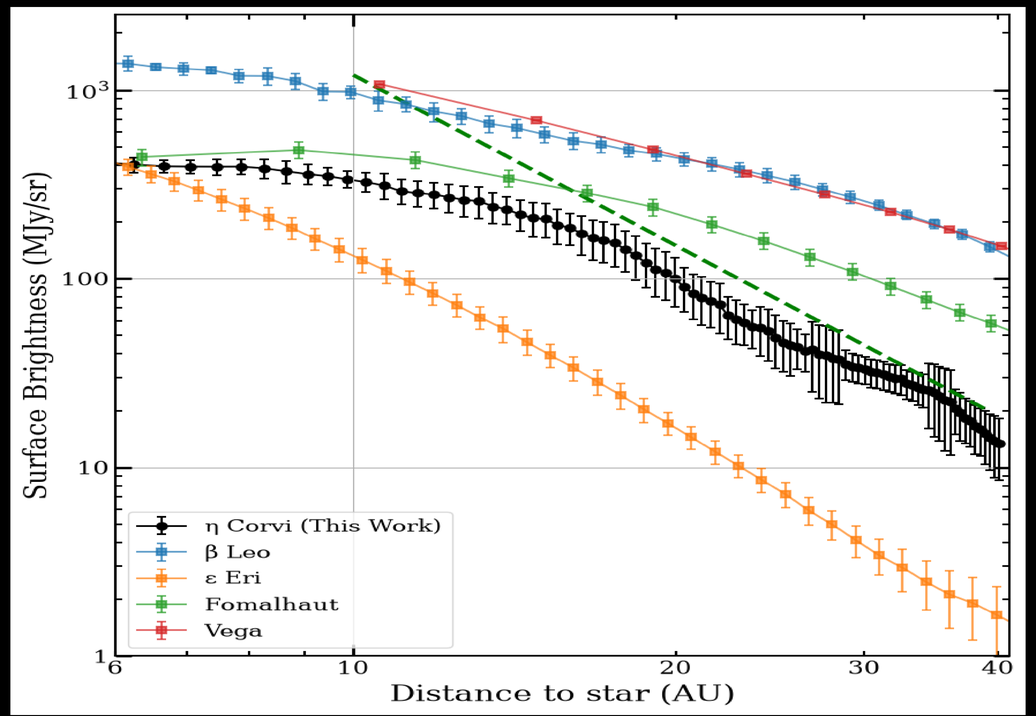
- **High-contrast Imaging:** Observation technique to detect faint objects (exoplanets, circumstellar disks, etc.) around bright stars.
- **Exozodiacal Dust:** analog to the zodiacal cloud - cosmic dust between planets in the Solar System.
- **About η Corvi** (Lebreton+ 2016)**:**
  - Spectral Type: F2V
  - Age: ~ 1.4 Gyr
  - Distance: ~ 58.7 light years
- **Why η Corvi:**
  - Consists of both warm dust and cold dust (Marino+ 2017)
  - Evidence for gas despite its old age
  - **Previous studies suggested that the warm dust was delivered into the inner region by comets.**

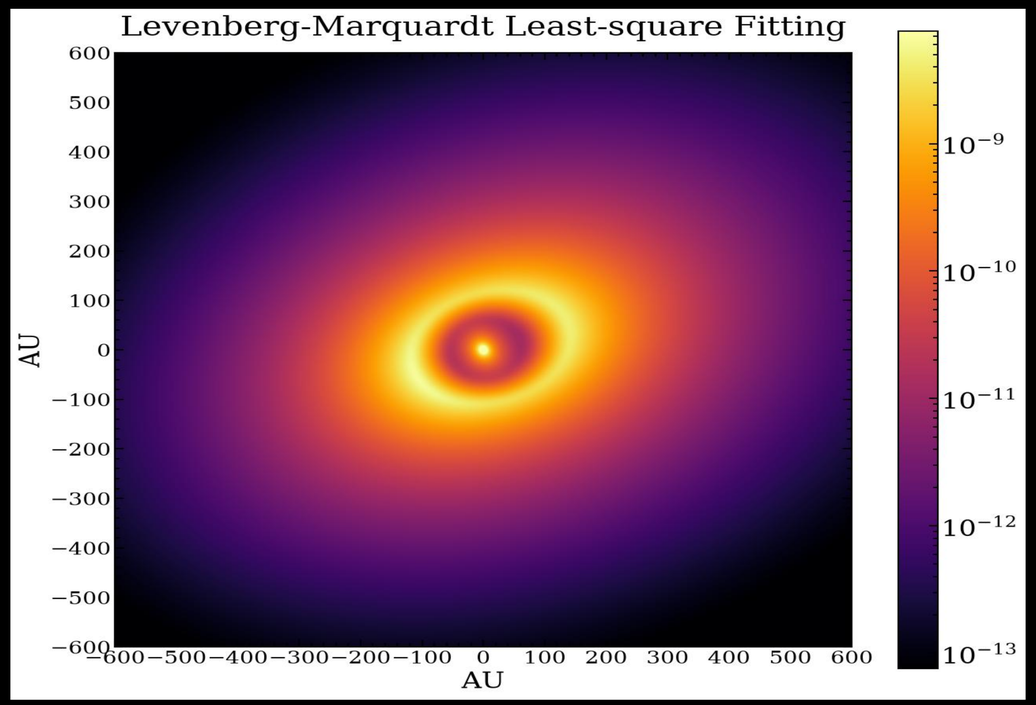
Methodology

- **Data Acquisition:** JWST GTO program #4538 – Exo-Asteroid Belts with JWST: Pinpointing the Warm Dust Emission
- **Data Analysis:**
  - Subtract the Point Spread Function (PSF) of the host star via Reference star Differential Imaging (RDI) with the Python package Winnie (Lawson+ 2022)
  - Problem: PSF subtraction also "over-subtracts" the real signals of the circumstellar disks
    Solution: Forward Modeling
    - Generate a synthetic model of the exozodi with known parameters
    - Project the disk model through the same subtraction matrix used on image data
    - Optimize the parameters using non-linear least square fitting until matching the PSF-subtracted image.
  - Extract disk geometry with Gaussian fitting
  - Extract radial brightness profile using concentric annuli



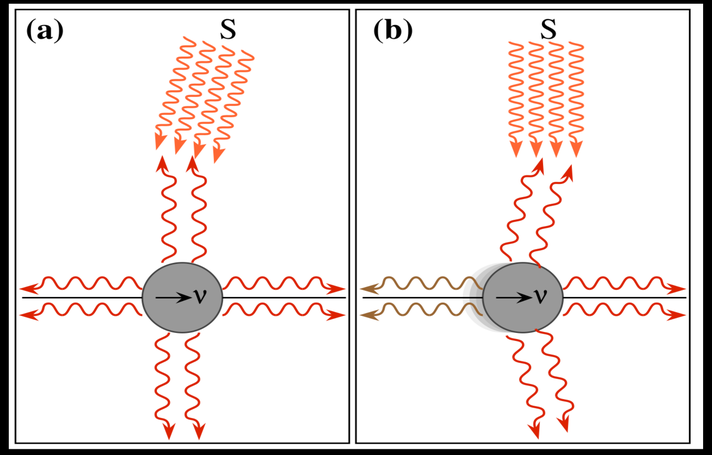
Plots





Results & Future Works

- Gaussian fits suggest a misalignment between inner exozodi and outer cold belt → Two components have different origins
- Disk parameters from Forward Modeling
- Slope of radial brightness profile suggests that the warm dust might have been dragged inwards by the host star's radiation under Poynting-Robertson effect.
- Future Implications:
  - Optimize the disk model parameters using maximum probability via the Monte Carlo Markov Chain (MCMC) method
  - Obtain spectra for the exozodi using MIRI Medium-Resolution Spectroscopy (MRS) to study its chemical composition and gain more insights into its origin.



References

- Lebreton, J., Beichman, C., Bryden, G., et al. 2016, ApJ, 817, 165
- Marino, S., Wyatt, M. C., Panic, O. et al. 2017, MNRAS, 465, 2595
- Lawson, K., Curie, T., Wisniewski, J. P. et al. 2022, ApJ, 935, 25

Acknowledgements

I would like to thank Dr. Schuyler Wolff for her guidance and support in completing this poster and throughout the project. I would like to thank the Galileo Circle Patron for your generous support during its progress.
The data presented in this poster were obtained from the Mikulski Archive for Space Telescopes (MAST) at the Space Telescope Science Institute. These observations are associated with JWST program 4538.

Contact Information

- Email: phuocnh21@arizona.edu
- LinkedIn: linkedin.com/in/phuoc-h-nguyen
- GitHub: github.com/thomasnguyen21

SCAN ME!